

INFORMATION VISUALISATION
(GEOM90007_2023_SM2 (Online))
ASSIGNMENT - III

GROUP 104

TEAM MEMBERS

NAME	ROLL NO	EMAIL
DIVYANSHU MISHRA	1281413	divyanshum@student.unimelb.edu.au
MEGHANA GANAPA	1436378	mganapa@student.unimelb.edu.au
SATVIK CHANDRA SAXENA	1414681	satvikchandra@student.unimelb.edu.au
AKASHDEEP SINGH	1387018	akashdeepsi2@student.unimelb.edu.au

Welcome to Melbourne Magic: Where Adventure Meets Culture!

Welcome to Melbourne, Australia's cultural capital, a city that seamlessly blends rich history, modern innovation, and stunning landscapes. Discover hidden laneway gems, immerse yourself in thrilling sports events, indulge in the world's finest coffee, and marvel at dynamic street art. With its unpredictable weather, Melbourne promises a unique adventure with four seasons in a day.

Join us as we unveil the secrets of this cosmopolitan metropolis. Explore iconic landmarks, diverse neighborhoods, and the soulful stories that make Melbourne a must-visit destination. Your Melbourne journey begins, where every moment is a masterpiece waiting to be discovered!

I. Design Summary

1. What the Interface Does:

The interface is a web-based tourist guide for the city of Melbourne, Australia. It provides visitors with comprehensive information about Melbourne's attractions, hotels, points of interest, cafes, restaurants, transportation routes and fun facts. Users can explore these categories through interactive tabs and access detailed information, images, and links to external sources. The "Nearby Attractions" section displays attractive destinations with images and descriptions. The "Fun Facts" section offers intriguing insights into Melbourne's culture and unique features. The "Overview" section gives users a brief introduction to the city.

2. How It Works:

The Melbourne Tourist Guide is a user-friendly Shiny web app that simplifies your exploration of the city. It features tabs for various aspects of Melbourne, including hotels, dining, transportation, overview, and fun facts. Each tab provides rich information, such as accommodation descriptions and images. The "Word Cloud" section offers dynamic visual representations, while the integration of Tableau in the last four tabs before the About Us tab provides interactive data visualizations for deeper insights.

This platform helps you uncover Melbourne's gems, make informed decisions, and immerse yourself in the city's culture. Join us on this virtual adventure to discover Melbourne's wonders through a blend of Shiny, content, visuals, and Tableau for an engaging experience.

3. Features of the Design for Which Credit is Sought:

User-Friendly Tabs: The use of tabs allows for a clean and organized interface, making it easy for users to navigate through various categories.

Images for Visual Appeal: Incorporating images of attractions in the "Nearby Attractions" section enhances visual appeal and helps users get a quick visual sense of the places.

Interactive Word Cloud: The "Word Cloud" section presents data in an engaging and visually interesting format. It allows users to see frequently mentioned terms at a glance.

Consistent Styling: The design maintains a consistent styling across the application. A gentle background image of Melbourne adds a touch of local flavor to the application.

4. Justification of Design Decisions:

Background Image: The background image of Melbourne helps set the tone for the tourist guide. It visually connects the user to the destination and conveys a sense of place.

Tab Organization: Using tabs for different sections is a design decision aimed at simplicity and user-friendliness. Visitors can quickly locate the information they are interested in without being overwhelmed by too much information on a single page.

Images for Attractions: Images are included in the "Nearby Attractions" section to provide visual context and attract the user's interest. Visual content often helps users make quicker decisions about places they'd like to visit.

Word Cloud Visualization: The word cloud is a visually appealing representation of frequently mentioned terms in the content. It provides an immediate snapshot of what's important or notable about Melbourne, which can pique the user's curiosity.

Color Scheme: The color scheme chosen (subtle gray and white) helps ensure good readability of the text while maintaining a modern and clean look. The colors are muted to ensure that the content and images take center stage.

Overall, the design prioritizes user experience, visual appeal, and simplicity. It aims to provide users with a visually engaging and informative experience to help them make the most of their visit to Melbourne.

About code:

The Tourist Guide of Melbourne relies on two main components: R Shiny and Tableau.

R Shiny is the backbone of the interface, serving as a web application framework for the R programming language. It provides the interactive structure, enabling users to navigate through Melbourne's information using tabs. Key libraries like ``shiny``, ``shinydashboard``, ``htmltools``, and ``wordcloud2`` enhance the app's functionality and appearance.

Tableau, a robust data visualization tool, is thoughtfully integrated into the interface, elevating data-driven insights in the last four tabs. It allows users to explore rich visualizations related to Melbourne's transportation networks and other relevant data. This integration offers a more engaging and informative experience.

These components, along with their associated libraries, work harmoniously to create a comprehensive and user-friendly platform, enriching the exploration of Melbourne's attractions and practicalities.

II. Summary

As a prospective visitor or a curious explorer of Melbourne, our Tourist Guide of Melbourne interface offers valuable insights and use cases for various user groups:

1. Exploring Melbourne's Iconic Landmarks (POI):

Use Case: Travelers can explore Melbourne's key attractions using the "POI" tab. It offers a visual map of POIs, each marked with a unique logo.

Discovery: User-friendly logos aid navigation. There are no external links, promoting in-person exploration. Find art, religious sites, architecture, or leisure spots with ease, each with detailed descriptions for a rich Melbourne experience.

2. Exploring Accommodation Options:

Use Case: Travelers seeking accommodations can explore the "Hotels" tab for a list of hotels with descriptions, images, and direct links for more information.

Discovery: Users can easily assess a range of options, from luxury to budget-friendly, and find accommodations that match their preferences.

3. Dining Choices for Food Enthusiasts:

Use Case: Food enthusiasts can dive into Melbourne's culinary scene through the "Cafes and Restaurants" tab, which provides comprehensive dining options with descriptions and external links.

Discovery: Users can explore diverse restaurants and cafes, aided by images and external links for menus and reviews, making dining decisions easier.

4. Efficient Transportation Planning:

Use Case: Travelers or locals seeking transportation routes can benefit from the "Transportation Routes" tab, with interactive maps and data visualizations thanks to Tableau integration.

Discovery: Users can visualize Melbourne's transportation network, identify key routes, and plan their journeys effectively with interactive maps.

Rationale:

Our user-centric interface offers detailed descriptions, images, and external links, ensuring efficient evaluation and informed decision-making for users. The combination of Tableau integration, interactive maps, and data visualizations enhances the depth of information available, aiding data-driven decision-making.

The visually engaging word cloud simplifies the identification of key attractions and cultural aspects of Melbourne, making content more accessible and memorable. Overall, our tool serves as a comprehensive guide, empowering users to make the most of their Melbourne experience, with a focus on efficiency, engagement, and informed exploration.

Melbourne Magic: Points of Interest (POIs) tab

In the Melbourne Tourist Guide, our user-friendly Points of Interest (POIs) filter allows you to personalize your city exploration. Simply choose a specific POI from the menu, and the map promptly adjusts to align with your interests. Whether you're passionate about exploring aquatic wonders at the Aquarium,

immersing yourself in Melbourne's rich culture at Art Galleries and Museums, uncovering the stories behind iconic Buildings, or delving into the diverse religious landscape of Churches, this feature ensures a personalized Melbourne experience.

Melbourne Magic: Restaurants and Cafes tab

The design is tailored to enhance users' exploration of Melbourne's cafes and restaurants, drawing from Victoria's open data. It offers a range of filtering options to cater to diverse preferences, such as cuisine type, vegetarian-friendliness, budget range, and pet or vegan-friendly establishments. This design aims to provide a user-centric experience, ensuring that users can navigate Melbourne's culinary landscape with ease and find the perfect restaurant to suit their individual tastes and requirements.

Melbourne Magic: Hotels tab

The "Hotels" tab within the Melbourne Tourist Guide equips users with robust filters for personalized accommodation searches, allowing customization based on star ratings, proximity to attractions, amenities, and pricing. This enhances the accommodation search experience. Complementing these filters, three impactful visualizations offer seamless hotel selection: the "Map with Hotel Logos by Star Rating" provides a visual map with categorized hotel logos, the "Table Showing Hotels by Price and Star Rating" facilitates comparisons of star ratings and pricing, and the "Smart Recommendation Graph" delivers personalized hotel recommendations, streamlining the selection process based on individual preferences and priorities.

Melbourne Magic: Transportation Routes Tab

The heart of this dashboard is the Interactive Transport Map, where each mode of transportation is easily distinguishable through distinct colors and logos. Tram Stops are marked with a unique logo and feature unique logos and display both the stop name and serviced tram numbers for quick identification, and Train Stations are represented by a distinctive emblem. Meanwhile, Bus Routes are depicted in their unique hue, giving users a visual representation of the city's extensive bus network, and hovering over a route reveals its origin and destination. Lastly, Bicycle Routes, with their distinct colors, reflect

Conclusion and Future Work

In the bustling heart of Melbourne, our Tableau-based Tourist Guide is your key to uncovering the city's hidden treasures. As you venture through this diverse metropolis, let the spirit of Melbourne guide your way.

Looking ahead, the Melbourne Tourist Guide has exciting potential. Integration of real-time data, user-generated content, augmented reality, and a range of features will enhance the user experience. Multilingual support, personalization, accessibility, and thematic expansion will make the guide even more inclusive. By staying responsive to user needs and embracing technological advancements, we aim to ensure that your Melbourne exploration is as dynamic and unforgettable as the city itself. So set forth, embrace every moment, and let the magic of Melbourne be your inspiration!

Data Sources:

Name	Link
Point of Interest	https://drive.google.com/file/d/1crnV653ZCGWhkBIzKo-mJCc9sY6ldBmN/view
Routes and stations	https://datashare.maps.vic.gov.au/search?q=uuid%3D871a5706-7282-5897-accd-cff576e71055/ https://metashare.maps.vic.gov.au/geonetwork/srv/api/records/0058eda4-7b1e-5681-9758-7adbda8481b1/formatters/sdm-html?root=html&output=html/
Train Stations:	https://datashare.maps.vic.gov.au/search?q=uuid%3D871a5706-7282-5897-accd-cff576e71055/ https://metashare.maps.vic.gov.au/geonetwork/srv/api/records/b6722101-8db5-51f0-8a6f-d1e4fe805b73/formatters/sdm-html?root=html&output=html
Hotels:	http://insideairbnb.com/get-the-data/
Restaurants and cafes	https://discover.data.vic.gov.au/dataset/cafe-restaurant-bistro-seats/resource/a3e8085b-1276-4695-8753-d2f9417ae66f/

Contribution:

Name	Contribution to project (<u>max 50 words</u>)	Percentage contribution
Divyanshu Mishra	In the project, I significantly developed and improved the R Shiny app's appearance aesthetically, applied design elements, and wrote a detailed report. I gathered data and also created the Tableau point of interest window and developed the overview tab and about us tab for an attractive and user-friendly experience. Made README file.	25%
Meghana Ganapa	I've developed a foundational R Shiny website, seamlessly integrating Tableau workbooks covering hotels, dining, routes, and points of interest. Additionally, I've created a detailed Tableau workbook focused on the hotel dataset. My contributions are outlined in the accompanying report and video presentation.	25%

Satvik Saxena	I collected vital datasets and sourced the main logo and images for R Shiny dashboard. I curated engaging location descriptions and developed a detailed Tableau workbook for routes and trams tab. My insights significantly enhanced the project's aesthetics, and I also contributed to the video presentation, ensuring clarity and engagement.	25%
Akashdeep	In this project I looked for the data sources which can be useful and integrated them to make into a more relevant dataset. Additionally, I created a well- defined tableau workbook which focused on the cafes and restaurants. Furthermore,I contributed to manage the report for the Restaurants and Cafes tab and video as well.	25%